

CrimeGPT

POLICE CASE DOCUMENTATION & LEGAL INTELLIGENCE PORTAL

Solution Architecture & Legal Code Transition Specifications

Designed for Indian Police Departments | Version 1.0.0

Executive Summary

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- **AI-Powered Legal Intelligence System:** engineered specifically to streamline the police investigation workflow from FIR registration to court production.
- **Transition Bridge:** Acts as a legal translation layer for officers migrating from colonial-era laws (IPC, CrPC, IEA) to modern codes (BNS, BNSS, BSA).
- **Form Digitization:** Incorporates scanned document OCR uploads, parsing text to populate case sheets under 10 seconds.
- **Forensic Transparency:** Maintains an interactive, chronological case diary timeline and logs compliance audit trails to ensure judicial credibility.

Indian Penal Reform Compliance

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- **BNS (Bharatiya Nyaya Sanhita):** replaced the colonial-era IPC (Indian Penal Code) enacted in 1860, updating punishment classifications.
- **BNSS (Bharatiya Nagarik Suraksha Sanhita):** replaced the CrPC (Code of Criminal Procedure) enacted in 1973, establishing new mandates for medical checkups, searches, and remand timings.
- **BSA (Bharatiya Sakshya Adhinyam):** replaced the IEA (Indian Evidence Act) enacted in 1872, detailing new guidelines for digital files and electronic forensics admissibility.
- **Solution Duty:** CrimeGPT serves as the operational translation portal, linking descriptions to correct legal acts under both frameworks concurrently.

Core Platform Features

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- **1. Scanned FIR OCR Uploads:** Drag-and-drop OCR scans that pre-populate form wizards.
- **2. AI Legal Translation Engine:** Suggests statutory sections and relevant landmark judgements.
- **3. Digital Case Diary:** Chronological timeline of statements, evidence recoveries, and arrests.
- **4. Multilingual Document Assembly:** Instantly drafts Remand, Medical, and Seizure Memos in English, Hindi, Gujarati, or Marathi.
- **5. Compliance Audit Logs:** Immutable admin trail tracking username details, actions, and parameters.

System Architecture & Tech Stack

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- • **Frontend:** Single Page React Application built with Vite and TypeScript. Styled using responsive TailwindCSS. State managers handle authentication headers and translation matrices.
- • **Backend:** Express.js Node API handling OCR processing, OpenAI API query assemblies, PDF compilation (PDFKit), and DOCX builds.
- • **Dual Database Layer:** PostgreSQL utilized in staging/production setups, with smart automatic fallback to SQLite (crimegpt.sqlite) for local network setups or zero-configuration offline field stations.
- • **Core Modules:** Tesseract.js for OCR, OpenAI GPT-4o-mini API for legal parsing, JWT for authentication.

Role-Based Access Control (RBAC)

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- **Investigating Officer (IO):** Registers case details, uploads OCR image templates, triggers AI legal advice, logs evidence, witness depositions, and outputs legal drafts.
- **Station House Officer (SHO):** Supervises station cases. Approves registrations, updates case status (ACTIVE, ARRESTED, CLOSED), and audits precinct stats.
- **Legal Advisor:** Read-only access to case timelines, summaries, and AI recommendations to review and prepare court charges.
- **System Administrator:** Governs IT settings and holds read-only clearance to the Compliance Audit Log to review modified values.

OCR & AI Legal Processing Pipeline

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- **Stage 1: Document Upload:** -> Scanned image or PDF of FIR uploaded -> OCR parses characters into strings.
- **Stage 2: Regex Extraction:** -> Pulls date values, name categories, and case/incident descriptions to populate database fields.
- **Stage 3: Narrative Suggestor:** -> Narrative details sent to OpenAI API model. If offline, runs a local regex keyword dictionary.
- **Stage 4: Legal Translation:** -> Mapped output generated containing relevant BNS, BNSS, BSA, and matching old IPC, CrPC, IEA sections with confidence scores and supreme court citations.

Digital Investigation Diary (Timeline)

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- **Statutory Requirement:** Section 172 of CrPC & corresponding BNSS sections mandate a diary detailing investigation steps day-by-day.
- **Automation Triggers:** CrimeGPT appends chronological timeline logs automatically on database operations (Filing case form, witness statements, weapon/evidence seizure, arrest booking, case closure).
- **Visual Timeline:** Provides judges and prosecutors an interactive checklist of events, ensuring custody chains and legal timelines remain transparent and valid.

Multilingual Document Generation

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- **Regional Localization:** Field operations require documents in regional state languages. CrimeGPT compiles templates in English, Hindi, Gujarati, and Marathi.
- **Dynamic Field Insertion:** Case database records are mapped directly into standard template letters.
- **Supported Legal Letters:** Remand Application (BNSS Sec 187), Medical Requisition (BNSS Sec 51/184), Seizure Memo (BNSS Sec 185), and Custody Jail Transfer request.

Legal Transition Reference Matrix

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Offense Type	Old IPC Sec	New BNS Sec	New BNSS (Procedural)	New BSA (Evidence)
Murder	Sec 300 / 302	Sec 101 / 103(1)	Inquest: Sec 176 BNSS	Dying Dec: Sec 32(1) BSA
Theft	Sec 378 / 380	Sec 303 / 305	Search: Sec 185 BNSS	Presumption: Sec 119 BSA
Hurt / Grievous	Sec 323 / 325	Sec 115 / 117	Medical: Sec 184 BNSS	Expert: Sec 45 BSA
Cheating / Forge	Sec 420 / 468	Sec 318 / 336	Property: Sec 105 BNSS	Electronic: Sec 63 BSA
Kidnapping / Ransom	Sec 359 / 364A	Sec 137 / 140	Statement: Sec 164 BNSS	Presumption: Sec 114 BSA
Rape / Modesty	Sec 375 / 376 / 354	Sec 63 / 64 / 74	Medical: Sec 184 BNSS	Consent: Sec 114A BSA

Compliance Audits & Integrity

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- **Immutable Action Trail:** Every creation, modification, status update, and download logs operator details in the audit database.
- **Audit Record Structure:** Tracks Officer codes, executed methods (CASE_REGISTERED, ARREST_MADE, etc.), timestamps, and JSON strings of changed parameters.
- **Security Enforcement:** Only accounts with the Administrator role can view audit trails. Logs cannot be modified or deleted through the user interface, enforcing security.

CrimeGPT

Empowering Digital Justice & Legal Compliance

- Accelerates police paperwork from hours to seconds.
- Guarantees legal transition accuracy between BNS & IPC.
- Restricts data visibility through role guards (RBAC).
- Secures evidence chain of custody via audit trails.
- Zero configuration setups ready out of the box.